

Detection Brainstorm

Assignment | 11/04/2026 | Face & Object Detection

01 5 Uses of Face / Object Detection

#	Use Case	Type	Description
01	Attendance System	Face	Automatically marks student/employee attendance by recognizing faces from a camera feed — no manual roll call needed.
02	Retail Shelf Monitoring	Object	Detects when products go out of stock on shelves and alerts staff to restock in real time.
03	Traffic Violation Detection	Object	Identifies vehicles jumping red lights, wrong-way driving, or speeding using road-mounted cameras.
04	Medical Imaging	Object	Detects tumors, fractures, or anomalies in X-rays and MRI scans to assist doctors in diagnosis.
05	Drowsiness Detection	Face	Monitors a driver's eyes and facial expressions to detect signs of fatigue and trigger an alert.

02 Designed Solution — Smart Attendance System

Problem Statement

Manual attendance in classrooms and offices is slow, error-prone, and can be manipulated through proxy attendance. A face-detection-based system automates this accurately and in a contactless manner.

How It Works — Pipeline

Step 1 Camera Feed	Step 2 Face Detection	Step 3 Face Recognition	Step 4 Confidence Check	Step 5 Log Attendance
Capture live video at entrance	OpenCV detects all faces in frame	Match against known faces DB	Threshold $\geq 80\%$ for valid match	Save Name + Date + Time to CSV

[Camera] → [Face Detection] → [Face Encoding] → [Match DB] → Mark PRESENT / Flag Unknown

Tools & Technologies

Component	Technology	Purpose
Language	Python 3.x	Core programming language
Detection	OpenCV + Haar Cascade / MTCNN	Detect faces in each frame
Recognition	face_recognition library	Encode and compare faces
Database	CSV file / SQLite	Store attendance records
Camera	Webcam / IP Camera	Capture live video feed

Component	Technology	Purpose
Alert	Email / SMS Notification	Notify for absentees

Advantages & Limitations

Advantages	Limitations
• Contactless and fully automatic • Works in real time • Cannot be manipulated by proxy • Generates instant digital records	• Accuracy drops in poor lighting • Requires initial enrollment phase • Privacy concerns — needs user consent

Real World Example: Used in Indian colleges and corporate offices with IP cameras at entry gates connected to an attendance management portal.